

# Yadynesh D Sonale

Chennai - 600127, Tamil Nadu, India  
+91-9766797915 • yadyneshsonale@gmail.com  
www.linkedin.com/in/yadynesh • www.github.com/LeNotaCoder



## Education

| Program                            | Institution                                                                        | %/CGPA  | Year    |
|------------------------------------|------------------------------------------------------------------------------------|---------|---------|
| B.Tech. (Computer Science & Engg.) | Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram | 8.87/10 | 2023-27 |

## Experience

1. NLP Research Intern: Chat Based Flight Recommender System  
(Supervisor: Dr. Vijay Daultani)

Aug 2025 - Present  
Neural Nurture

  - Surveyed recent literature from ICLR/EMNLP/ACL on prompt optimization and in-context learning; implemented and reproduced state-of-the-art methods (OPRO, MIPRO, BootstrapFewShot) using DSPy for a flight recommendation agent.
  - Conducted systematic experiments with LLaMA and GPT models, evaluating prompt strategies using ranking metrics such as Precision@K, MRR, and nDCG to analyze and compare optimizer performance.
  - Skills: In-Context Learning (ICL), SoTA Prompting Techniques, DSPy, Feature Engineering
2. Research Intern: Multimodal Fundus + OCT Disease Classification  
(Guide: Dr. Preeth R)

May-Jul 2025  
Hybrid Intelligence Lab

  - Developed a multimodal deep learning pipeline integrating fundus and OCT images using transfer learning with DASNet and FineFeatureCNN, followed by DualStreamNet fusion, achieving 98.87% binary and 97.45% three-class DR classification accuracy.
  - Implemented Fibonacci-based preprocessing for OCT texture enhancement and carried out model training, ablation studies, and comparative analysis.
2. Research Intern: Fundus Image Classification for Diabetic Retinopathy  
(Guide: Dr. Preeth R)

Dec 2024-Apr 2025  
Dept. of Computer Science

  - Introduced DASNet, a dual-branch CNN integrating MaxPooling, Adaptive MaxPooling, and Spatial Pyramid Pooling to capture subtle and prominent retinal features, achieving 95.34%–97.46% accuracy across BiDR, APTOS, and EDI datasets, outperforming DenseNet121, VGG16, ResNet50, and InceptionV3 by 1.85%–11.28%.
  - Designed HSV+CLAHE preprocessing pipeline to enhance contrast in fundus images under variable lighting, boosting DASNet accuracy by 1.52%–2.10% and enabling robust early-stage DR detection for telemedicine applications.
2. Research Intern: Breast Abnormality Classification using Thermal Images  
(Guide: Dr. Avinash Kumar)

Sep 2024-Jan 2025  
Dept. of Mechanical Engg.

  - Achieved 96.36% accuracy and 100% recall in detecting possibly malignant breast abnormalities using fine-tuned ResNet-50 on 238 segmented thermal images from 119 patients, presented at ASME IMECE India 2025.
  - Boosted model performance from 45% to 96% by implementing manual ROI segmentation, preprocessing, and data augmentation to address noise and class imbalance in thermal imaging data.

## Publications

- Yadynesh D Sonale, et al. "DASNet: A Dual Adaptive Subtle-Feature Network for Enhanced Diabetic Retinopathy Detection in Fundus Images." Accepted at International Conference on Pattern Recognition Applications and Methods (ICPRAM), 2026.
- Yadynesh D Sonale, et al. "Deep learning-based classification of breast abnormalities using thermal imaging and ResNet-50." In Proceedings of the ASME International Mechanical Engineering Congress & Exposition (IMECE), 2025.

- **Yadynesh D Sonale**, et al. "*A comparative study of pre-trained deep learning models with and without pre-processing for multi-class classification of thermal breast images in early cancer detection.*" In Proceedings of the **ASME International Mechanical Engineering Congress & Exposition (IMECE)**, 2025.

## Projects

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### 1. *Hour Long Video Question Answering*

Aug-Sep 2025

*ICCV Perception Challenge 2025*

Google Deepmind

- Created an **LSTM-based temporal modeling pipeline** for **video question answering**, integrating **CLIP embeddings** to handle hour-long multimodal video data efficiently.
- Designed a **frame compression algorithm** leveraging temporal similarity metrics to minimize redundancy while preserving essential sequential information for improved model performance.

### 2. *Motor Imagery EEG Classification with CNN-LSTM Hybrid & EEGNet*

CS5109

*(Faculty: Dr. Kannadasan K)*

Course: Brain Computer Interfaces

- Implemented a **transfer learning-based CNN-LSTM hybrid** model using **Inception-v3** and **CWT** on BCI Competition IV 2a, achieving **92% accuracy** in 4-class motor imagery classification.
- Benchmarked the hybrid model against **EEGNet**, providing comparative analysis of performance, efficiency, and generalization.

### 3. *Stock Market Analysis*

CS3006

*(Faculty: Dr. Preeth R)*

Course: Introduction to Data Science for Engineers

- Built a stock analysis pipeline using **yfinance**, performing data cleaning, feature engineering, and technical indicators like **RSI** and **MACD**.
- Developed ML models (**Logistic Regression**, **Linear Regression**, **Random Forest**) to predict price movements with visual evaluation of predictions and feature importance.

## Technical Skills

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- **Languages:** C, C++, Bash, Python
- **Tools:** Latex, Git, Docker, Microsoft Office
- **Libraries:** PyTorch, TensorFlow, Numpy, Pandas, Matplotlib, Scikit-Learn, Snsrcape, DSPy, Langchain, Hugging Face

## Volunteering and Extracurricular Activities

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- **Member** of the Institute Basketball Team.
- **Team Lead & Senior Consultant** at PiVOT Consulting (Aug 2024 – Aug 2025).
- **Captain** for HUSTLERS Team in the Fadeaway Intra-College Basketball Tournament.
- **Event Coordinator** at Vashisht 2024 Organized hackathons conducted by the Google Developer Student Club.

## Achievements/Awards

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- Among the **Youngest Teams** to Reach the **Final Round** of the **Toyota CODE FOR SAFER INDIA Hackathon 2019**.
- **98.64 percentile** in **JEE-MAINS** among 1.7 million candidates to pursue Bachelors at IIITDM Kancheepuram
- Achieved **91.30 percentile** in **JEE Advanced**, placing among the top performers nationwide in India's premier engineering entrance examination.

## References

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- [Dr. Vijay Daultani](#)
- [Prof. Preeth R](#)
- [Prof. Avinash Kumar](#)

*Founder & CEO, Neural Nurture*  
*Assistant Professor, IIITDM Kancheepuram*  
*Assistant Professor, IIITDM Kancheepuram*